
Temperature Scaling in Discrete Sequence (Language) Models

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Abstract

Temperature scaling (TS) is widely used to control generation behavior in discrete sequence models. However, TS at the sequence level is challenging as the needed partition function is typically intractable and naive *token-wise* TS induces bias. Discrete diffusion settings exacerbate this issue: enhanced flexibility due to multiple potential orderings for generating tokens complicates likelihood computation and incurs high variance. We propose and investigate fine-tuning objectives for sequence-level TS, along with metrics to quantify the success of a given TS procedure. Specifically, our contributions are threefold. First, we introduce our objective across a wide class of sequence models through a unifying formalism. Second, we design two metrics that quantify TS based on likelihood *ratios* and thus obviate access to the partition function. Finally, we introduce fine-tuning objectives that reliably achieve the desired change in model temperature. Our experiments with language models (GPT2, BD3LM) show that the proposed approach leads to more consistent generation with lower Negative Log-Likelihood (NLL). Furthermore, we provide empirical evidence that it can enhance the reasoning performance in language models. Code is available at github.com/Aalto-QuML/temp_lm.

1. Introduction

Generative sequence models are widely used in modern machine learning. They solve generation problems by factorizing over subproblems, i.e., tokens, allowing them to tackle complex tasks such as language processing.

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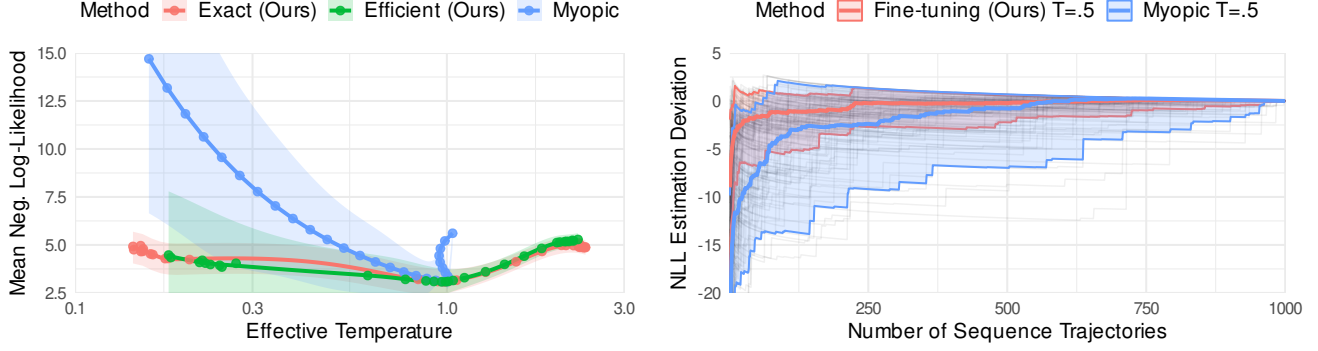
Table 1. Overview of the main contributions in this work.

Different models perform this factorization in different ways: autoregressive models, the current state-of-the-art method, do so sequentially (Radford et al., 2019; Brown et al., 2020; Vaswani et al., 2017), while discrete diffusion models (Austin et al., 2021; Campbell et al., 2022; Lou et al., 2024; Sahoo et al., 2024; Shi et al., 2024; Ou et al., 2025; Nie et al., 2025; Ye et al., 2025) operate in a random order. Hybrid methods that combine some aspects of autoregressive and discrete diffusion have also been proposed (Sahoo et al., 2026; Arriola et al., 2025; Wang et al., 2026; Wu et al., 2026b;a; Song et al., 2025).

A standard approach to modulate the behavior of any model, including sequence models, is *temperature scaling* (TS). The idea of temperature scaling is to adjust the sharpness of the model distribution, e.g., to improve performance in downstream f or control generation diversity. More formally, one can anneal the base model distribution p by a scalar $T \in \mathbb{R}_{>0}$, so that the resulting model $p_T \propto p^{1/T}$. For $T < 1$, the distribution is sharper, pushing probability mass toward higher-likelihood outputs; conversely, for $T > 1$, the distribution is flatter, yielding a higher diversity of outputs.

While temperature scaling is intuitive and commonly used, subtle caveats are often overlooked in its motivation and application. In order to avoid this, we first discuss: *Why do we want to scale temperatures in generative modeling?*

For some ambient space $\mathcal{X}$, most generative tasks have a notion of *good* elements that we would like to sample from. The common assumption is that model likelihoods correlate well with sample quality, so if we can increase the average model likelihood, we presumably increase the



(a) Comparison between our proposed methods and token-wise (myopic) temperature scaling. Models were biased towards different temperatures. We show mean NLL (lower is better) and variance-indicating ribbons. Fine-tuning is able to scale to low *and* high temperatures with reduced performance loss and lower variance.

(b) Running estimated NLL for 64 different sequences with an increasing amount of unmasking trajectories. NLL estimates are highly variable, even with many sampled trajectories. Fine-tuning shows drastically lower variance than myopic temperature scaling.

Figure 1. Comparison of our procedure versus token-wise temperature scaling with BD3LM (Arriola et al., 2025), measured on the OpenWebText dataset (Gokaslan & Cohen, 2019).

average quality as well. For example, when we assume there exists a set of uniformly *good*, high-density elements $A \subset \mathcal{X}$ in the space $\mathcal{X}$ of all possible outputs, the goal of temperature scaling perfectly aligns with the question “how can we increase $p(A)$?” For any temperature-scaled distribution $p_T = p^{1/T}(x)/Z_{p_T}$ with $0 < T < 1$ and $Z_{p_T} = \sum p^{1/T}$ as the partition function, we have that

$$p(x) > p(x') \iff \frac{p_T(x)}{p_T(x')} > \frac{p(x)}{p(x')}.$$

For the high-density set A , we then have $p_T(A) > p(A)$.

Inconveniently, approximating p_T from p is non-trivial in *sequence models* since the partition function Z_{p_T} is intractable even under moderate sequence lengths. Nevertheless, there are still many methods that aim to control the likelihoods of sequence models (Joy et al., 2023; Ding et al., 2021; Shen et al., 2024; Choi et al., 2024; Shih et al., 2023). For example, Karan & Du (2026) observe that low-temperature sampling can improve performance on reasoning tasks.

Traditionally, temperature scaling is instead done at the token level, scaling the probability of individual tokens without considering the sequence as a whole. This simple approach can lead to miscalibrated sequence distributions, causing problems such as overconfidence or sequence degeneration (Guo et al., 2017; Holtzman et al., 2020). This instability is illustrated in the blue curve of Figure 1a, which shows a decrease in performance for $T \neq 1$.

When we try to learn a sequence model $q_T \propto p_T$, the right objective is non-trivial, as available techniques are restricted by the intractable partition function Z_{p_T} . Furthermore, to scale the likelihood of a sequence, the likelihood must first be well-defined. However, in most sequence-generation models, the likelihood of a sequence is not uniquely deter-

mined by the model parameters—it depends on the parameters of the sampling process itself. In most state-of-the-art discrete diffusion models (Sahoo et al., 2024; Shi et al., 2024; Ou et al., 2025; Nie et al., 2025), the order of generation itself is subject to randomness, so the same sequence can be sampled in many different ways. This leads to three issues specific to stochastic generation ordering.

- Models exhibit very high variance in their estimated likelihoods for a given sequence, depending on the trajectory of the generation process, meaning that simply fixing this order biases the model’s estimates. This effect is illustrated in Figure 1b, which shows the drastic variability of order-dependent likelihood estimates.
- Discrete diffusion language models do not restrict generation orders, which leads to a combinatorial explosion for computing expected likelihood over all trajectories. For this reason, discrete diffusion models commonly only use a lower bound on the likelihood.
- Sequence models follow a variety of different formalisms. Some rely on discrete or continuous time and define noising processes (Austin et al., 2021; Campbell et al., 2022) while others use order-dependent formalisms (Radford et al., 2019), and parallel generation of multiple tokens may or may not be allowed.

Techniques for sequence-model temperature scaling do exist: among them, LHTS (Shih et al., 2023) directly targets the temperature-scaled distribution for autoregressive models, while more general distribution-matching approaches based on GFlowNets (Hu et al., 2024; Venkatraman et al., 2024) can in principle recover temperature scaling by choosing the tempered model likelihood as their reward. In contrast

to this, [Karan & Du \(2026\)](#) provide an inference-only algorithm that relies on heavy resampling.

In this work, we investigate sequence-level temperature scaling specifically for stochastic-order sequence models under one unified formalism. We develop methods for computing and approximating their marginalized sequence likelihoods, introduce partition-free metrics for measuring sequence-level temperature, and derive offline fine-tuning objectives that avoid repeated on-policy sampling. More specifically, we adapt the KL-minimizing loss of [Shih et al. \(2023\)](#) to stochastic-order sequence models, and propose two well-behaved objectives based on likelihood ratios, one exact and one faster estimate.

We evaluate our method on block diffusion ([Arriola et al., 2025](#)) and GPT2 ([Radford et al., 2019](#)). We show that this scheme (i) achieves effective temperature scaling at the sequence level and (ii) has better performance than token TS. Furthermore, we show that our method can lead to better results on reasoning benchmarks for Phi-3.5-mini-instruct ([Abdin et al., 2024](#)) and Qwen2.5-7B ([Yang et al., 2024](#)). An overview of the structure of this paper and our contributions can be seen in Table 1.

2. Unified Notation for Language Models

Notation for language models varies across literature and assigns different semantics to the “order of generation”. In this work, we present a general and lightweight formalism that unifies generative sequence models. We denote sequences of length L over a given vocabulary $\mathcal{X}$ with $x \in \mathcal{X}^L$, and for an index $i \leq L$ denote the i -th token with $x_i \in \mathcal{X}$, for a given index-set $I \subset \{1, \dots, L\}$ denote the subsequence $(x_i \mid i \in I)$ with $x_I \in \mathcal{X}^{|I|}$, and the *prefix* $(x_1, \dots, x_{i-1})$ with $x_{<i} \in \mathcal{X}^{i-1}$. For integers $n_1, n_2 \in \mathbb{N}, n_1 \leq n_2$, we further denote with $[n_1] = \{1, \dots, n_1\}$ and with $[n_1, n_2] = \{n_1, \dots, n_2\}$. A generative sequence model generates sequences according to a (possibly stochastic) *unmasking schedule*.

Definition 2.1 (Unmasking Schedule). Let $x \in \mathcal{X}^L$ be a sequence over the vocabulary $\mathcal{X}$ of length L . We then define a K -step unmasking schedule π as

$$\pi : [0, K] \rightarrow \mathcal{P}([L])$$

where $\pi_0 = \emptyset$ and $\pi_K = [L]$. We restrict π to be strictly monotone, i.e., $\forall s, t \in [0, K]$ with $s < t$ it holds that $\pi_s \subsetneq \pi_t$. π_t is then the set of positions whose token values are fixed after step t .

We then use x_{π_t} for the partially unmasked sequence containing the tokens at positions in π_t , and denote the token indices unmasked at step t with $\Delta_t(\pi) := \pi_t \setminus \pi_{t-1}$. For the rest of the paper, we use the shorthand $\Delta_t = \Delta_t(\pi)$, unless ambiguous. For a given model p , we denote the

probability of a sequence x being generated by p under π , similarly to [Kim et al. \(2025\)](#) as

$$p(x; \pi) := \prod_{t=1}^K p(x_{\Delta_t} \mid x_{\pi_{t-1}}). \quad (1)$$

When the unmasking schedule is stochastic, we denote it by the random variable Π . With this definition, we can characterize the following model classes solely in terms of different distributions over Π , i.e., by their different strategies for unmasking tokens and, consequently, by different possible factorizations of $p(x)$. As with deterministic schedules, we write Π_t for the set of positions unmasked after step t of a stochastic schedule Π .

3. Background

With our canonical notation, we can now define typical model types for sequence generation in a standardized, interoperable way.

3.1. Autoregressive Models

Autoregressive models are the state of the art for sequence generation, particularly in language generation ([Radford et al., 2019](#); [Brown et al., 2020](#); [Vaswani et al., 2017](#); [Zheng et al., 2025](#)). For any autoregressive model p , the probability of a given sequence $x \in \mathcal{X}^L$ factorizes as a product of conditionals over prefixes. To express index-based autoregression as a schedule, we define π^{AR} with $\pi_t^{\text{AR}} = [t]$ and $\Delta_t = \{t\}$.

$$p(x; \pi^{\text{AR}}) = \prod_{t=1}^L p(x_{\Delta_t} \mid x_{\pi_{t-1}^{\text{AR}}}) = \prod_{i=1}^L p(x_i \mid x_{<i}). \quad (2)$$

3.2. Masked Diffusion

Masked diffusion models (MDMs), as described in most literature, define a forward noising process over discrete or continuous time that stochastically masks token positions ([Austin et al., 2021](#); [Campbell et al., 2022](#); [Lou et al., 2024](#); [Sahoo et al., 2024](#); [Shi et al., 2024](#); [Ou et al., 2025](#)). The MDM is then trained to reverse this time-dependent process. Time dependence in MDMs is realized *exclusively* via the noising schedule $\alpha : [0, 1] \rightarrow [0, 1]$. The schedule α defines the unmasking probability for a masked token at time τ , i.i.d. across indices. For this paper, we define an MDM to be any model where Π^α follows an unrestricted distribution according to some i.i.d. schedule α .

MDMs are trained using a negative ELBO to maximize the expected log-likelihood of sequences in a dataset, uniformly over unmasking *permutations* ([Uria et al., 2014](#); [Ou et al., 2025](#); [Kim et al., 2025](#))

$$\mathcal{L}_{\text{diff}} = -\mathbb{E}_{x \sim \mathcal{D}, \Pi \sim \mathcal{U}(S_L)} \left[\sum_{t=1}^K \log p(x_{\Delta_t} \mid x_{\Pi_{t-1}}) \right], \quad (3)$$

where $\mathcal{U}(S_L)$ denotes a uniform distribution over permutations of length L . At sampling time, MDMs might unmask multiple tokens at the same time t , where the conditional independence of all tokens in Δ_t is assumed, i.e.,

$$p(x_{\Delta_t} | x_{\pi_{t-1}}) = \prod_{i \in \Delta_t} p(x_i | x_{\pi_{t-1}}).$$

This *parallel unmasking* improves sampling efficiency by reducing the effective number of sampling steps. However, the independence assumption introduces a distribution shift and might drastically decrease model performance (Kang et al., 2026; Chang et al., 2022; Austin et al., 2021).

3.3. Block Diffusion

Block diffusion models (Arriola et al., 2025; Wu et al., 2026b;a; Wang et al., 2026) are a restricted class of MDMs in which sequences are partitioned into (fixed-length) *blocks* of length $B \ll L$. In block diffusion, any unmasking schedule *within* a given block is allowed, but different blocks are generated *in sequence*. More formally, the stochastic unmasking schedule Π^B in block diffusion only has support on *block-respecting* π . We call π block-respecting if it can be defined as a concatenation of block schedules π^{B_i} such that

$$p(x; \pi) = \prod_i p(x_{B_i} | x_{B_{<i}}; \pi^{B_i}). \quad (4)$$

This hybrid approach increases flexibility in sequence length and enables drastically more efficient training.

3.4. Temperature Scaling

A central technique for controlling the trade-off between diversity and quality in sequence models is *temperature scaling* (Guo et al., 2017; Holtzman et al., 2020; Desai & Durrett, 2020; Li et al., 2025). Given a density p over some space $\mathcal{X}$ we want to approximate for some fixed temperature $T \in \mathbb{R}_+$ the annealed distribution p_T where

$$p_T(x) = p(x)^{1/T} / Z_{p_T}. \quad (5)$$

Here $Z_{p_T} = \sum p(x)^{1/T}$ is the normalizing partition function of p_T . Renormalization is required for temperature scaling, but for many (non-sequence) model classes, this can be done by logit scaling, allowing for trivial one-parameter temperature scaling (Guo et al., 2017).

Naive Temperature Scaling on Sequence Models When the domain $\mathcal{X}$ is small, computing Z_{p_T} is non-problematic. However, for language models, the vocabulary of *each token* already contains thousands of logits. Computing Z_{p_T} explicitly for sequences of length L requires normalization over $|\mathcal{X}|^L$ possible outputs and is completely intractable.

Shih et al. (2023) show that token-wise, or *myopic* temperature scaling fails on sequences. Denoting for some

sequence x with $Z_{p_T}^{\Delta_i} = \int p(x_{\Delta_i} | x_{\pi_{i-1}})^{1/T}$ the tractable computable token-wise partition functions, it holds that

$$p_T(x; \pi) \propto \frac{\prod_i p(x_{\Delta_i} | x_{\pi_{i-1}})^{1/T}}{Z_{p_T}} \propto \prod_i \frac{p(x_{\Delta_i} | x_{\pi_{i-1}})^{1/T}}{Z_{p_T}^{\Delta_i}}.$$

This mismatch makes the naive approach to temperature scaling inapplicable for sequence models, and introduces known pathologies (Holtzman et al., 2020).

Long Horizon Temperature Scaling Shih et al. (2023)

tackle sequence-level temperature scaling for autoregressive models with the fine-tuning procedure *long horizon temperature scaling* (LHTS). LHTS trains the temperature-scaled sequence model q_T to approximate the annealed fixed model p_T with likelihood maximization under importance sampling. Formally, LHTS minimizes

$$\begin{aligned} \mathcal{L}_{\text{LHTS}} &= \mathbb{E}_{x \sim p} [-w_T(x_i) \log q_T(x_i | x_{<i})] \\ w_T(x_i) &= \exp \left(\frac{1-T}{T} \log p(x | x_{<i}) - \log Z_{p_T} \right). \end{aligned} \quad (6)$$

Here Z_{p_T} is constant and does not need to be computed. $\mathcal{L}_{\text{LHTS}}$ can be shown to minimize the KL-divergence $\text{KL}(p_T \| q_T)$. However, for sequence models, the formulation is only applied to autoregressive models. Furthermore, the variance of the weights w_T demands a careful training regimen to manage stability, biasing the objective (Shih et al., 2023).

4. Stochastic Sequence Unmasking Models

To apply TS to a given model, we need a well-defined notion of the likelihood of a sequence. This is nontrivial in stochastic settings, where the same sequence can be generated by different schedules, and the probability across schedules can vary widely, as shown in Figure 1b.

In this section, we define the class of stochastic sequence unmasking models (SSUMs). This general class is compatible with a wide body of the existing literature on autoregressive *as well as* discrete diffusion models. We then present an approach based on dynamic programming to compute the likelihood of the sequence under a given model, conditioned on its scheduling distribution.

4.1. Stochastic Sequence Unmasking Models

Different sequence models are usually viewed through distinct formalisms. We provide a unified perspective for many models, viewing progressive unmasking of tokens as their common task (albeit according to different schedules). In contrast to other unified formalisms (Fathi et al.,

Model	Sampling Method	Distributions	Models
Parallel unmasking MDM	any order, parallel	Π^α	Absorbing D3PM (Austin et al., 2021), MDLM (Sahoo et al., 2024), MD4 (Shi et al., 2024), RADD (Ou et al., 2025)
Sequential unmasking MDM	any order, sequential	$\mathcal{U}(S_L)$	same as above
Autoregressive Model	In order, sequential	$\delta_{\pi^{\text{AR}}}$	GPT (Radford et al., 2019)
Block diffusion	block-respecting order	$\Pi^{B_i, \alpha}, \mathcal{U}(S_B^M)$	BD3LM (Arriola et al., 2025)

Table 2. Overview of models, sampling methods, distributions, and references.

2025; Hoogetboom et al., 2022), our formulation is time-independent yet can describe time-dependent MDMs.

Definition 4.1 (Stochastic Sequence Unmasking Model).

Let $p(x_i \mid x_S)$ be a sequence model that predicts a token x_i at position $i \notin S$, conditioned on an observed partial sequence x_S . Let Π be a random variable over unmasking schedules. We say p is a *stochastic sequence unmasking model* (SSUM) if for all π with $\Pr(\Pi = \pi) > 0$:

- π is strictly monotone, i.e., each position is unmasked exactly once (no remasking).
- $p(x_i \mid x_S)$ depends only on i and x_S .
- Π is independent of the token values x .

Autoregressive models are SSUMs as $\Pr(\Pi = \pi^{\text{AR}}) = 1$ and therefore $\Pr(\Pi \neq \pi^{\text{AR}}) = 0$ and π^{AR} satisfies the SSUM conditions. Furthermore, many MDMs are SSUMs as the following proposition shows.

Proposition 4.2. *Absorbing D3PM (Austin et al., 2021), MDLM (Sahoo et al., 2024), MD4 (Shi et al., 2024), RADD (Ou et al., 2025) are SSUMs.*

Remark 4.3 (Dependent SSUMs). We can relax the independence condition to prefix-dependent schedules $\pi_t(x_{\pi_{t-1}})$. This extension of our definition allows us to describe diffusion models with confidence- and entropy-based decoders (Nie et al., 2025; Ye et al., 2025) as (dependent) SSUMs. Many of the following results can be stated equivalently for dependent SSUMs.

Calculating the probability under a model p can be difficult, as the probability using different unmasking schedules π, π' might not be the same $p(x; \pi) \neq p(x; \pi')$ and for longer sequences, the variance between different schedules can vary significantly (Uria et al., 2014; Ou et al., 2025). Instead, the expectation over the unmasking schedules can be used to compute the probabilities.

Remark 4.4. Let p be a SSUM, then the probability of a sequence $x \in \mathcal{X}^L$ can be calculated by marginalizing over the unmasking schedules π

$$p(x) = \mathbb{E}_\Pi[p(x; \Pi)].$$

Corollary 4.5. *Absorbing D3PM, MDLM, MD4, RADD over continuous time unmask according to $\Pi \sim \mathcal{U}(S_L)$, i.e., a sequence $x \in \mathcal{X}^L$ is unmasked sequentially and*

$$\Pr(x) = \mathbb{E}_{\Pi \sim \mathcal{U}(S_L)}[p(x; \Pi)] = \frac{1}{L!} \sum_{\pi} p(x; \pi). \quad (7)$$

As autoregressive models and MDLMs are SSUMs, it follows that block diffusion models are also SSUMs, as they are built on these models and change only the distribution of Π , which is in continuous time $\mathcal{U}(S_B^M)$, i.e., $\Pr(\pi) = \frac{1}{B!^M}$.

While we can compute the marginal log-likelihood, it unfortunately does not admit an efficient unbiased approximation:

Remark 4.6 (No fixed-sample unbiased estimator for the log marginal likelihood). For fixed x, n , and arbitrary Π and p satisfying $0 < p(x; \pi) < 1$, there is no universal unbiased fixed finite-sample estimator of

$$\log p(x) = \log \mathbb{E}_\Pi[p(x; \Pi)].$$

In the following section, we discuss methods to compute $\log p(x)$ as efficiently as possible, and later utilize a biased Monte-Carlo estimator for $\log p(x)$.

4.2. Exact Computation of Sequence Likelihoods

For our fine-tuning objective, we now need to be able to compute $p(x)$ for a given SSUM p and a given sequence $x \in \mathcal{X}^L$. Unfortunately, in the general case, the support of Π is exponential in L for MDMs. Allowing for parallel unmasking, the number of possible unmasking schedules is equal to the number of different *weak orders* on L elements, also known as the *ordered Bell numbers*, which are in $\mathcal{O}(L!1.5^L)$. We can, however, drastically improve upon this pessimistic bound in *block diffusion* with *continuous time*.

Proposition 4.7. *Let p be a block diffusion model, with block size B , in a sequential unmasking regime. Then, for a given x , we can compute $p(x)$ in time $\mathcal{O}(L2^B)$.*

This faster computation can be achieved by using a dynamic program that marginalizes over partial sequences rather than over unmasking sequences. For block diffusion, we

can calculate the probability of each partial block $x_{\pi_t^{B_i}}$ using the probabilities $p(x \mid x_{\pi_{t+1}^{B_i}})$ in the following way

$$p(x \mid x_{\pi_t}) = \frac{1}{B-t} \sum_{j \in B_i \setminus \pi_t} p(x \mid x_{\pi_t \cup \{j\}}) p(x_j \mid x_{\pi_t}) \quad (8)$$

starting with $p(x \mid x_{B_i}) = 1$. This computes the probability of each partial $x_{\pi_t^{B_i}}$ once resulting in a runtime of $\mathcal{O}(L2^B)$. Algorithm 1 gives the dynamic program for block diffusion.

Algorithm 1 Dynamic program for computing $p(x)$ exactly

```

1: Input: Block diffusion model  $p$ , sequence  $x \in \mathcal{X}^L$ 
2: Output:  $p(x)$ 
3: for each block  $B_i$  do
4:    $p(x_{B_i} \mid x_{<B_i}, x_{\pi_B^{B_i}}) = 1$ 
5:   for  $t = B-1, \dots, 0$  do
6:     for each partial schedule state  $\pi_t^{B_i}$  do
7:        $p(x_{B_i} \mid x_{<B_i}, x_{\pi_t^{B_i}}) = \dots \{\text{Equation (8)}\}$ 
8:     end for
9:   end for
10:   $p(x_{B_i} \mid x_{<B_i}) = p(x_{B_i} \mid x_{<B_i}, x_{\pi_0^{B_i}})$ 
11: end for
12:  $p(x) = \prod_i p(x_{B_i} \mid x_{<B_i})$ 
    
```

5. Stochastic Sequence Temperature Scaling

In this section, we introduce Stochastic Sequence Temperature Scaling (SSTS) as a technique for fine-tuning a model’s sequence-level temperature. As an initial contribution, we introduce metrics to validate a model’s sequence-level temperature (SST) without access to its partition function. Building on these fundamental concepts, we introduce three distinct objective functions for SSTS.

5.1. Partition-Free Temperature Scaling Metrics

As established in Section 1, temperature scaling does not change the likelihood ranking of p :

$$p(x) > p(x') \iff q_T(x) > q_T(x')$$

To measure the agreement between the rankings induced by q_T and p on some dataset of sequences $\mathcal{D}$, we use Kendall’s τ , a standard measure of rank (Kendall, 1938),

$$\tau = \frac{1}{\binom{|\mathcal{D}|}{2}} \sum_{\{x, x'\} \subset \mathcal{D}} \text{sgn} \left(\log \frac{p(x)}{p(x')} \right) \text{sgn} \left(\log \frac{q_T(x)}{q_T(x')} \right). \quad (9)$$

Because the logarithm is monotonic, the rankings are preserved in log space. Moreover, taking log-ratios cancels the partition functions Z_p and Z_{q_T} , so unnormalized models are sufficient. The rankings induced by p and q_T perfectly agree when $\tau = 1$, and completely disagree when $\tau = -1$.

Rank-preservation alone is an incomplete metric for temperature scaling. Inconveniently, the precise numeric relation between p and q_T is influenced by the unknown partition function Z_{p_T} . This unknown, however, cancels when we investigate likelihood *ratios*

$$q_T \propto p^{1/T} \Rightarrow \frac{q_T(x)}{q_T(x')} = \left(\frac{p(x)}{p(x')} \right)^{1/T} = \frac{p^{1/T}(x)/Z_{p_T}}{p^{1/T}(x')/Z_{p_T}}.$$

We utilize this observation and refine our objective as

Lemma 5.1 (Temperature Scaling via Likelihood Ratios). *Let p, q_T be probability densities with global support on some space $\mathcal{X}$. It holds that $q_T \propto p^{1/T}$ iff*

$$\forall x, x' \in \mathcal{X}^L : \log \frac{p(x)}{p(x')} = T \log \frac{q_T(x)}{q_T(x')}. \quad (10)$$

For two models p, q_T , define their *temperature log-ratio* for a given sequence x as

$$h(x) := \log p(x) - T \log q_T(x) \quad (11)$$

and try to fine-tune a model q_T using our procedure such that $\forall x : h(x) \approx c$ for some constant c , while keeping the variance of the ratio across sequences low. To estimate the resulting temperature change, we can use linear regression on the log-likelihoods.

Definition 5.2 (Effective Temperature). Let p, q_T be probability densities over some space $\mathcal{X}$, then we estimate the *effective temperature* of q_T on a dataset $\mathcal{D}$ with homogeneous least squares regression:

$$T_{\text{eff}} := \frac{\sum_{x, x' \in \mathcal{D}} \log \frac{p(x)}{p(x')} \log \frac{q_T(x)}{q_T(x')}}{\sum_{x, x' \in \mathcal{D}} \left(\log \frac{q_T(x)}{q_T(x')} \right)^2} \quad (12)$$

5.2. Fine-Tuning Objectives for SSTS

After defining temperature ratios as a metric for the effectiveness of TS, we now extend sequence-level temperature-scaling loss functions to any SSUM, i.e., any model that adheres to Definition 4.1. During this section, we define p to be a fixed pre-trained SSUM and compute $p(x)$ according to Section 4.2 as well as all relevant conditional probabilities over sub-sequences. We investigate the following three distinct temperature-scaling objectives with complementary characteristics.

5.2.1. IMPORTANCE SAMPLING SSTS

To extend LHTS in Equation (6) to the stochastic unmasking schedules of SSUMs, we apply importance weighting to the standard token-wise training loss, averaged over unmasking schedules. This gives the SSTS objective

$$\mathcal{L}_{\text{SSTS}} = - \mathbb{E}_{\substack{\pi \sim \Pi \\ x \sim p}} \left[\sum_{t=1}^K w_T(x_{\pi_{t-1}}) \log q_T(x_{\Delta_t} \mid x_{\pi_{t-1}}) \right] \quad (13)$$

with weights and baselines defined as

$$w_T(x_{\pi_t}) = \exp\left(\frac{1-T}{T} \log p(x | x_{\pi_t}) - \mu(x_{\pi_t})\right)$$

$$\mu(x_{\pi_t}) = \mathbb{E}_{x \sim p(\cdot | x_{\pi_t})} \left[\frac{1-T}{T} \log p(x | x_{\pi_t}) \right].$$

The conditional probability for a given partial schedule π_t^* can be calculated as in Section 4.2 and is defined as

$$p(x | x_{\pi_t^*}) := \mathbb{E}_{\Pi} [p(x | x_{\pi_t^*}; \Pi) | \Pi_{t'} = \pi_{t'}^* \forall t' \leq t].$$

Under mild conditions, we can show that $\mathcal{L}_{\text{SSTS}}$ is a proper loss function for our objective.

Proposition 5.3. *Let p be a pre-trained SSUM with global support and let $T > 0$. Assume that p is path-consistent, i.e., $\forall \pi \in \text{supp}(\Pi) : p(x | x_{\pi_t}, \Pi_{0:t} = \pi_{0:t}) = p(x | x_{\pi_t})$. If μ is finite everywhere, then $\mathcal{L}_{\text{SSTS}}$ is strictly proper for p_T , i.e., it minimizes $\text{KL}(p_T \| q_T)$.*

This importance-sampling-based approach allows us to define a loss function that minimizes the KL divergence (Cover & Thomas, 2006) to our target distribution; however, the exponential weights w exhibit high variance in practice due to the inherently high variance of $\log p(x | x_{\pi_t})$. In their paper, Shih et al. (2023) utilize a variety of variance-reducing techniques to stabilize training. Among these techniques, they resort to clipping high weights. This, or similar, approaches are hard to avoid when using importance sampling, but they introduce bias into the objective (Elvira et al., 2019; Korba & Portier, 2022). To mitigate these weaknesses, we propose a regularization scheme that minimizes h 's variance and drastically stabilizes training.

5.2.2. ENTROPY VARIANCE REGULARIZATION

The loss function $\mathcal{L}_{\text{SSTS}}$ we introduced in Section 5.2.1 is strictly proper, but unstable in practice. $\mathcal{L}_{\text{SSTS}}$ can be equivalently expressed as minimizing the expected temperature log-ratio $\mathbb{E}_{p_T}[h]$. To stabilize training, we propose a regularization scheme that encourages h to be constant, i.e., minimizing $\text{Var}_{p_T}[h]$ (Hayashi, 2009). We do this with regression towards the equality in Equation (10), and define our *ratio regularizer* as

$$\mathcal{L}_{\text{ratio}} = \mathbb{E}_{x, x' \sim p_T} \left[\left(\log \frac{p(x)}{q_T(x)^T} - \log \frac{p(x')}{q_T(x')^T} \right)^2 \right] \quad (14)$$

$$= \mathbb{E}_{x, x' \sim p_T} \left[(h(x) - h(x'))^2 \right] \quad (15)$$

This log-variance objective is closely related to the log-variance loss underlying *VarGrad* (Richter et al., 2020; Richter & Berner, 2024) and to squared log-ratio objectives such as *Trajectory Balance* (TB) (Malkin et al., 2022).

Richter et al. (2020) show that, after differentiating the log-variance loss and choosing the current variational distribution as reference, its gradient recovers the KL gradient. TB instead applies a squared log-ratio constraint to individual trajectories and learns the corresponding normalization constant.

In our setting, $\mathcal{L}_{\text{ratio}}$ is instead defined directly on the marginalized sequence likelihoods $p(x) = \mathbb{E}_{\Pi}[p(x; \Pi)]$ and $q_T(x) = \mathbb{E}_{\Pi}[q_T(x; \Pi)]$. Furthermore, in our setting the reference distribution is the fixed p_T , rather than the evolving q_T . This deliberate choice allows us to implement a variation of this objective with much more efficient sampling, but changes its relation to KL minimization:

Proposition 5.4. *Let p be a pretrained SSUM with global support over $\mathcal{X}^L$, and let $T \in \mathbb{R}_+$ be the target temperature. A model q_T that minimizes $\mathcal{L}_{\text{ratio}}$ minimizes the relative entropy variance $\text{Var}(p_T \| q_T)$. When $\text{Var}(p_T \| q_T) = 0$, it holds that $\text{KL}(p_T \| q_T) = 0$.*

Unfortunately, Proposition 5.4 does not guarantee that a small nonzero relative entropy variance implies a small KL divergence. Intuitively, the main issue is *likelihood drift*, where q_T might assign arbitrarily small likelihoods to sequences with high probability under p_T . We avoid drift by using $\mathcal{L}_{\text{ratio}}$ as a *regularizer*, combined with the default (token-wise) likelihood maximization objective in Equation (3).

The next issue to address in $\mathcal{L}_{\text{ratio}}$ is, as in $\mathcal{L}_{\text{SSTS}}$, importance sampling. Equation (15) is defined over p_T , which we cannot directly sample from. In order to enable efficient sampling, we desire an objective over the base distribution p . However, exact change of measure from p_T to p would require importance weights $w(x)w(x')$, where

$$w(x) = \exp\left(\frac{1-T}{T} \log p(x) - \log Z_{p_T}\right).$$

Instead of approximating these importance weights, we deliberately change the reference distribution of our log-variance loss from p_T to p without importance weighting. This change of reference distribution yields an efficient offline regularizer $\mathcal{L}_p = \text{Var}_p[h]$ with the same unique target distribution as $\mathcal{L}_{\text{ratio}}$ under realizability:

$$\mathcal{L}_p = 0 \iff \text{Var}_p \left[\log \frac{p_T(x)}{q_T(x)} \right] = 0 \iff q_T = p_T.$$

We therefore use $\mathcal{L}_p$ as a practical offline objective. In practice, samples from the data distribution $\mathcal{D}$ may be used directly in place of samples from p .

5.2.3. MONTE-CARLO REGULARIZATION

In the previous section, we proposed two partition-free regularizers $\mathcal{L}_{\text{ratio}}$ and $\mathcal{L}_p$, the latter being able to target

p_T just with samples from p and without unstable importance weighting. Both objectives enforce that the marginal $q_T(x) = \mathbb{E}_\Pi[q_T(x; \pi)] \propto p^{1/T}(x)$. However, even with Algorithm 1, the computational cost grows exponentially with block size (Proposition 4.7). Further, it is also nontrivial to estimate $p(x)$ well (Remark 4.6), so we cannot avoid the exponential compute cost without biasing our estimate. To nevertheless provide a scalable alternative to $\mathcal{L}_{\text{ratio}}$ and $\mathcal{L}_p$, we use the k -sample Monte-Carlo estimator

$$\log \tilde{p}(x) := \log \left[\frac{1}{k} \sum_{i=1, \pi^{(i)} \sim \Pi}^k p(x; \pi^{(i)}) \right], \quad (16)$$

and define $\log \tilde{q}_T(x)$ analogously. For $k \rightarrow \infty$, clearly $\log \tilde{p} \rightarrow \log p$, so the choice of k controls the admitted bias of our estimator. We then define our MC objective as

$$\mathcal{L}_k = \mathbb{E}_{x, x' \sim p} \left[\left(\log \frac{\tilde{p}(x)}{\tilde{q}_T(x)^T} - \log \frac{\tilde{p}(x')}{\tilde{q}_T(x')^T} \right)^2 \right], \quad (17)$$

where for $k \rightarrow \infty$, we have $\mathcal{L}_k = \mathcal{L}_p$. On the other hand, for the case $k = 1$, when the sampled schedules are the same, we recover the *path-/schedule-wise* variant of VarGrad (Richter et al., 2020; Richter & Berner, 2024). In practice, it can be useful to choose different values k for $\log \tilde{p}$ and $\log \tilde{q}_T$, as outputs of p can be precomputed, allowing larger batches.

6. Experiments

In this section, we demonstrate the effectiveness of our approach in language modeling and aim to answer the following questions empirically:

- Which *actual* target temperatures T can we achieve depending on model performance?
- Are our variance-based objectives influenced by *likelihood drift* without the default SSUM objective $\mathcal{L}_{\text{diff}}$?
- For block diffusion, how does our procedure influence the schedule variance? How does exact computation of $p(x)$ perform compared to likelihood estimates in $\mathcal{L}_k$?
- How is downstream task performance, like reasoning, influenced by our fine-tuning procedure?

Setup To address these questions, we compare our fine-tuned models against myopic temperature scaling for block diffusion and autoregressive models. To compare our metrics, we fine-tune the pretrained BD3LM from Arriola et al. (2025) and GPT2 (Radford et al., 2019) on the OpenWebText corpus (Gokaslan & Cohen, 2019) using either $\mathcal{L}_p$, $\mathcal{L}_k$ or just myopic scaling. We target extreme temperatures

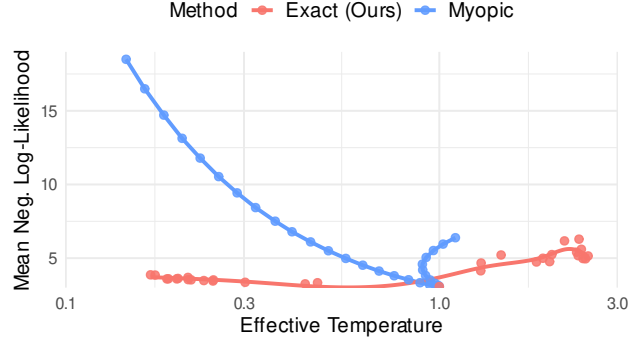


Figure 2. Comparison between $\mathcal{L}_p$ (ours) and token-wise (myopic) temperature scaling with GPT2 measured on the OpenWebText dataset. Models were biased towards the temperatures 0.1 and 3.0. We show mean NLL (lower is better). Fine-tuning reaches low and high temperatures with reduced performance loss.

$T = 0.1$ and $T = 3$ to investigate the achievable range and to assess the calibration of our procedure and myopic scaling at a temperature of $T = 0.8$. All models are trained on a dataset of 30000 sequences on an A100-80G GPU.

In a second experiment, we investigate how this procedure affects downstream task performance. For this, we use the two autoregressive language models Qwen2.5-7B (Yang et al., 2024) and Phi-3.5-mini-instruct (Abdin et al., 2024). We fine-tune both models on the MATH dataset (Hendrycks et al., 2021) and then evaluate accuracy on MATH500 (Lightman et al., 2024) and HumanEval (Chen et al., 2021). For all our experiments, we observe T_{eff} (Definition 5.2), τ (Equation (9)) as well as sequence-NLL on an OWT test set. For the diffusion experiments, we also observe the variance of $\log p(x; \pi)$ over different schedules.

6.1. Results

Effective Temperature Range We investigate how much our approach can change model behavior with extreme temperature targets. The results of this task are visualized in Figures 1a and 2 and Table 4. Myopic TS (baseline) effectively sharpens the distributions, reaching effective temperatures close to 0.1, but interestingly, does not achieve a *flatter* distribution with $T_{\text{eff}} > 1$. Furthermore, for diffusion, low temperatures lead to a drastic increase in schedule variance. In comparison, both $\mathcal{L}_k$ and $\mathcal{L}_p$ achieve better effective temperature ranges, with much lower NLL, and do not blow up the schedule variance. The choice of k shows that it influences variance; however, NLL is low even for $k = 1$, indicating that this loss is highly feasible with the biased objective. Surprisingly, while $\mathcal{L}_p$ reduces schedule variance most, it also significantly reduces τ . The results for GPT2 exhibit the same pattern.

Table 3. Results on MATH500 and HumanEval. The shown values are pass@1 accuracies, averaged over 5 seeds. We include a run with $T = 1$ as an ablation study to test if the sequence-level objective improves model performance without actual TS.

Model	Dataset	Base	Myopic	$\mathcal{L}_p, T = 1$ (ours)	$\mathcal{L}_p, T = 0.25$ (ours)
Phi-3.5-mini-instruct	MATH500	0.330	<u>0.446</u>	0.417	0.452
	HumanEval	0.066	0.035	0.101	<u>0.095</u>
Qwen2.5-7B	MATH500	0.214	0.376	0.419	<u>0.399</u>
	HumanEval	0.296	0.539	0.318	<u>0.370</u>

Model	Name	$T_{\text{eff}(\text{min-max})}$	NLL ₅	Var ₅	τ_5
BD3LM	Myopic	0.158–1.04	5.28	624.3	0.862
	$\mathcal{L}_{k=1}$	0.178–2.25	3.41	134.6	0.851
	$\mathcal{L}_{k=4}$	0.162–2.24	3.39	118.0	0.856
	$\mathcal{L}_p$	0.143–2.36	4.24	72.3	0.750
GPT2	Myopic	0.145–1.10	5.50	—	0.888
	$\mathcal{L}_p$	0.169–2.50	3.33	—	0.803

Table 4. Results of effective temperature range. One model was fine-tuned for low/high temperature, respectively, and intermediate checkpoints were used to produce the results. The objective was combined with the SSUM objective $\mathcal{L}_{\text{diff}}$. T_{eff} shows the effective minimum and maximum temperature ratios. NLL is normalized over the sequence length, variance is averaged over log-likelihoods.

Target Temperature Precision We investigate how precisely a specific temperature can be achieved to test the calibration of our approach. Table 5 presents the results. Across the different models, $\mathcal{L}_p$ generally changes the effective temperature most. The token-wise scaling does not seem to translate directly to changes at the sequence level, and the KL-based objectives $\mathcal{L}_{\text{SSTS}}$ and LHTS (Shih et al., 2023) only achieve minor changes in temperature. The effect of using $\mathcal{L}_{\text{diff}}$ is uniform: it prevents likelihood drift and lowers NLL, but it also reduces the effect of temperature scaling. As in Table 4, the choice of k leads to improved approximation of $\mathcal{L}_p$, but seems to influence the NLL less.

Reasoning As a final experiment, we investigate reasoning tasks as an application of our method: We apply our fine-tuning procedure to more modern reasoning models and evaluate their downstream performance on MATH500 and HumanEval. Table 3 shows the results. The effect is greatly model-dependent: Phi-3.5 benefits more from fine-tuning than myopic TS in both tasks. This is surprisingly more pronounced in HumanEval, which is not well-aligned with the chosen fine-tuning dataset. For Qwen2.5, this is different: Myopic TS greatly boosts performance, especially on HumanEval. In summary, the effect of either technique on downstream tasks varies widely across models and tasks.

Model	Name	$\mathcal{L}_{\text{diff}}$	T_{eff}	NLL	Variance	τ
BD3LM	Myopic		0.93	3.15	70.15	0.964
	$\mathcal{L}_{\text{SSTS}}$		0.97	3.04	<u>46.39</u>	0.958
	$\mathcal{L}_{k=1}$	×	0.85	3.23	69.76	0.948
		✓	0.91	3.11	56.85	0.955
	$\mathcal{L}_{k=4}$	×	0.84	3.25	81.82	0.948
		✓	0.90	3.11	60.58	<u>0.961</u>
	$\mathcal{L}_p$	×	0.80	3.44	86.73	0.945
		✓	0.93	<u>3.07</u>	40.06	<u>0.960</u>
GPT2	Myopic		0.94	3.19	—	0.953
	LHTS		0.89	3.10	—	0.809
	$\mathcal{L}_p$	×	0.86	3.29	—	0.927
		✓	0.90	<u>3.11</u>	—	<u>0.936</u>

Table 5. Results of calibration experiment losses. The models were fine-tuned with $T = 0.8$. Myopic temperature scaling is used as the baseline. NLL is normalized over the sequence length. Variance is averaged over the log-likelihoods.

7. Conclusion

In this work, we present a unifying view on TS for sequence generation models. The key ingredients for both enforcing and quantifying TS are a well-defined notion of sequence likelihood when generation schedules are stochastic and a tractable and objective metric. With these two formalisms, we define multiple objectives for temperature scaling.

Remarkably, our likelihood-ratio-based objectives $\mathcal{L}_p$ and $\mathcal{L}_k$ show much more stable performance in practice compared to vanilla importance sampling and myopic temperature scaling. Future work could investigate principled guarantees and approximations for sequence likelihood in parallel settings. Future work could also further investigate the effect of TS on downstream tasks such as molecular generation.

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Impact Statement

This paper presents work whose goal is to advance the field of machine learning. There are many potential societal consequences of our work, none of which we feel must be specifically highlighted here.

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A. Proofs

Proposition 4.2. *Absorbing D3PM (Austin et al., 2021), MDLM (Sahoo et al., 2024), MD4 (Shi et al., 2024), RADD (Ou et al., 2025) are SSUMs.*

Proof. When showing that the models are SSUMs, we need to show that they fulfill the following conditions for all π with $\Pr(\Pi = \pi) > 0$:

1. π is strictly monotone, i.e., each position is unmasked exactly once (no remasking).
2. $p(x_i \mid x_S)$ depends only on i and x_S .
3. Π is independent of the token values x .

We show this for MDLMs; for the others it follows analogously.

First, let Π be a random variable describing the distribution of the MDLM unmasking schedule. By definition, once a token is unmasked, it cannot be changed. It follows that π is strictly monotone for all π with $\Pr(\Pi = \pi) > 0$.

For condition (2), the denoising model itself is time-independent: time dependence enters only through the noising schedule α , which determines the distribution of masked positions, while the model predicts the clean-data conditional associated with the observed partial sequence (Sahoo et al., 2024; Ou et al., 2025). Consequently, conditioned on a fixed partial sequence x_S , the distribution predicted for position i depends only on i and x_S , and can therefore be written as $p(x_i \mid x_S)$.

For condition (3), the noising schedule α induces i.i.d. random unmasking times τ_i for the token positions. Their distribution depends only on α and not on the token values x . Mapping these unmasking times to the corresponding unmasking schedule therefore induces a distribution Π^α that is independent of x . □

Corollary 4.5. *Absorbing D3PM, MDLM, MD4, RADD over continuous time unmask according to $\Pi \sim \mathcal{U}(S_L)$, i.e., a sequence $x \in \mathcal{X}^L$ is unmasked sequentially and*

$$\Pr(x) = \mathbb{E}_{\Pi \sim \mathcal{U}(S_L)}[p(x; \Pi)] = \frac{1}{L!} \sum_{\pi} p(x; \pi). \quad (7)$$

Proof. For continuous time, the probability of unmasking two tokens at the same time is zero, as it is equivalent to sampling discrete points (time steps) $t_1, t_2 \sim \mathcal{U}[0, 1]$ which is a null set. This means that for all π only one token get unmasked at a time, so the only thing distinguishing the π is in which order the tokens are getting unmasked. As $\Pr(\pi)$ is independent of time and position, it is the same for all π . Giving us a uniform distribution over the symmetric group $\Pi \sim \mathcal{U}(S_L)$. □

Proposition 4.7. *Let p be a block diffusion model, with block size B , in a sequential unmasking regime. Then, for a given x , we can compute $p(x)$ in time $O(L2^B)$.*

Proof. We assume we can choose $M \in \mathbb{N}$ such that $L = MB$. For each token position $i \in [L]$, there are 2^{B-1} possible partial sequences x_S when i is unmasked. This means $p(x)$ can be written in terms of exactly $L2^{B-1}$ unique factors $p(x_i \mid x_S)$, each of which requires one forward pass through the model p . Then $p(x)$ can be computed using the dynamic programming method illustrated in Algorithm 1. □

Proposition 5.3. *Let p be a pre-trained SSUM with global support and let $T > 0$. Assume that p is path-consistent, i.e., $\forall \pi \in \text{supp}(\Pi) : p(x \mid x_{\pi_t}, \Pi_{0:t} = \pi_{0:t}) = p(x \mid x_{\pi_t})$. If μ is finite everywhere, then $\mathcal{L}_{\text{SSTS}}$ is strictly proper for p_T , i.e., it minimizes $\text{KL}(p_T \parallel q_T)$.*

Proof. Fix a schedule π , a step t , and a partial sequence $x_{\pi_{t-1}}$. Up to the positive factor $\exp(-\mu(x_{\pi_{t-1}}))$, the corresponding term in $\mathcal{L}_{\text{SSTS}}$ is

$$-\mathbb{E}_{x \sim p(\cdot \mid x_{\pi_{t-1}})} \left[p(x \mid x_{\pi_{t-1}})^{\frac{1-T}{T}} \log q_T(x_{\Delta_t} \mid x_{\pi_{t-1}}) \right]. \quad (18)$$

Since

$$p(x \mid x_{\pi_{t-1}}) p(x \mid x_{\pi_{t-1}})^{\frac{1-T}{T}} = p(x \mid x_{\pi_{t-1}})^{1/T} \propto p_T(x \mid x_{\pi_{t-1}}), \quad (19)$$

this is, up to a positive constant, the conditional cross-entropy under p_T . Hence it is strictly proper and minimized by $q_T(x_{\Delta_t} \mid x_{\pi_{t-1}}) = p_T(x_{\Delta_t} \mid x_{\pi_{t-1}})$. Averaging these strictly proper losses over steps and schedules preserves the same unique optimum, yielding $q_T = p_T$ and therefore minimizing $\text{KL}(p_T \parallel q_T)$:

$$\mathcal{L}_{\text{SSTS}} = -\mathbb{E}_{\pi \sim \Pi} \left[\sum_{t=1}^K \mathbb{E}_{x_{\pi_{t-1}} \sim p} \left[\mathbb{E}_{x \sim p(\cdot \mid x_{\pi_{t-1}})} [w_T(x_{\pi_{t-1}}) \log q_T(x_{\Delta_t} \mid x_{\pi_{t-1}})] \right] \right]. \quad (20)$$

□

Proposition 5.4. *Let p be a pretrained SSUM with global support over $\mathcal{X}^L$, and let $T \in \mathbb{R}_+$ be the target temperature. A model q_T that minimizes $\mathcal{L}_{\text{ratio}}$ minimizes the relative entropy variance $\text{Var}(p_T \parallel q_T)$. When $\text{Var}(p_T \parallel q_T) = 0$, it holds that $\text{KL}(p_T \parallel q_T) = 0$.*

Proof. We define $h(x) = \log p(x) - T \log q_T(x)$. Then it holds that

$$\log \frac{p(x)}{p(x')} - T \log \frac{q_T(x)}{q_T(x')} = \log p(x) - \log p(x') - T \log q_T(x) + T \log q_T(x') \quad (21)$$

$$= \log p(x) - T \log q_T(x) - (\log p(x') - T \log q_T(x')) \quad (22)$$

$$= h(x) - h(x') \quad (23)$$

Consequently, $\mathcal{L}_{\text{ratio}} = \mathbb{E}_{x, x' \sim p_T} [(h(x) - h(x'))^2] = 2\text{Var}_{p_T} h(x)$. Per definition of h , it holds with convention of Hayashi (2009)

$$\log \frac{p_T(x)}{q_T(x)} = \frac{1}{T} h(x) - \log Z_{p_T} \implies \text{Var}_{p_T} h(x) = T^2 \text{Var}(p_T \parallel q_T) := T^2 \text{Var}_{p_T} \left[\log \frac{p_T(x)}{q_T(x)} \right]. \quad (24)$$

Now it directly holds that $\mathcal{L}_{\text{ratio}} = 2\text{Var}_{p_T} h(x) = 2T^2 \text{Var}(p_T \parallel q_T)$, which proves the first claim for $T > 0$.

Now, if $\text{Var}(p_T \parallel q_T) = 0$, then $\log \frac{p_T(x)}{q_T(x)} = c$ for some constant c on the support of p_T . Hence $p_T(x) = e^c q_T(x)$. Since both p_T and q_T are normalized, $e^c = 1$, and therefore $p_T = q_T$. It follows that $\text{KL}(p_T \parallel q_T) = 0$.

□

B. Experiments

Code is available at github.com/Aalto-QuML/temp_lm with all details for reproducibility.